UDAY KUMAR

ECE Undergraduate | RF & Antenna Research | IEEE Student Branch President

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PROFESSIONAL SUMMARY

ECE undergraduate with hands-on RF and microwave antenna research experience at CARE (Centre for Applied Research in Electronics), IIT Delhi, including analytical design and full-wave EM simulation in CST Studio Suite of X-band microstrip patch antennas and slotted waveguide arrays, with parametric analysis of a 10 GHz microstrip patch antenna. Currently developing a slot-loaded microstrip patch antenna for bandwidth enhancement, with a manuscript in preparation. Recipient of the Best Paper Award at IEEE IECS 2026 and Regional Finalist at NXP Cup India 2026, with additional experience in embedded systems, PCB design, ROS2-based autonomous systems and IoT development.

RESEARCH & INTERNSHIP EXPERIENCE

RF & Microwave Antenna Research Intern — Indian Institute of Technology Delhi (IIT Delhi)

New Delhi, India June 2026 – July 2026

- Designed and simulated high-gain slotted waveguide and microstrip patch antennas for X-band applications using CST Studio Suite, analyzing antenna gain, bandwidth, and return loss.
- Carried out full-cycle antenna performance analysis (S-parameters, gain optimization, return loss, bandwidth and radiation pattern characterization) under direct faculty supervision.
- Contributed to active wireless communication research, feeding into a co-authored manuscript currently in preparation.

PCB Design Intern — Manav Rachna International Institute of Research and Studies

Faridabad, India June 2025 -- July 2025

- Executed end-to-end PCB development — schematic capture, component placement, high-speed routing, and DRC verification using Altium Designer.
- Delivered production-ready PCB layouts for embedded hardware modules, ensuring signal integrity and manufacturability compliance.

EDUCATION

B.Tech – Electronics & Communication Engineering, Manav Rachna International Institute of Research and Studies

Delhi NCR, India 2024 – 2028

SGPA: S1 – 7.35 | S2 – 8.10 | S3 – 8.25 | S4 – 8.73 (Upward Academic Trajectory)

PUBLICATIONS & MANUSCRIPTS

- "IoT and Sensor Fusion in Smart Helmets: A Review Toward Safer and Sustainable Mobility", IEEE IECS 2026, VIT Bhopal University .
- "Design and Performance Analysis of a Slot-Loaded Microstrip Patch Antenna for Enhanced Bandwidth." Presented at the IETE Zonal Seminar and ISF Congress 2026, MITS-DU, Gwalior, India, 22–23 August 2026.

HONORS & ACHIEVEMENTS

Best Paper Award — IEEE International Conference on Intelligent Electronics & Computational Systems (IECS 2026)

IEEE IECS 2026 (1st edition), VIT Bhopal University

- Presented the review paper "IoT and Sensor Fusion in Smart Helmets: A Review Toward Safer and Sustainable Mobility" at the conference.
- Received the Best Paper Award for contribution to IoT-enabled safety systems and sustainable mobility research.

NXP Cup India 2026 – Regional Finalist | Top 100 of 4,600+ Teams Nationwide

NXP Semiconductors India 2026

- Advanced to the Regional Finals after qualifying among the Top 100 teams from 4,600+ competing teams across India — a top 2.2% placement nationally.
- Engineered a ROS2-based autonomous vehicle featuring real-time computer vision, sensor fusion, line following, and obstacle avoidance; integrated multi-sensor systems for robust autonomous navigation under competition conditions.
- Contributed to software integration, autonomous system testing, and performance optimization, enabling successful qualification for the Regional Finals.

LEADERSHIP

IEEE Student Branch President — Manav Rachna International Institute of Research and Studies

- Elected to lead the university IEEE chapter; organizing technical workshops and industry collaboration events across 100+ student members.
- Active IEEE member and technical hackathon participant.

KEY PROJECTS

PixiePlot – AI-Powered Storytelling Web App (Ongoing) www.pixieplot.com

Building an AI-powered storytelling web application using Next.js, React, TypeScript, and Firebase for personalized children's story generation. Implemented user authentication, story management, voice narration, responsive design, and PWA functionality, with a focus on scalability, performance, accessibility, and SEO best practices.

Uday Tour & Travels Website uday-travels-in.vercel.app

Developed and deployed a responsive website for a real-world travel business, showcasing tour packages, vehicle rental services and customer inquiries, hosted on Vercel.

Smart Helmet – Accident Detection & Alert System (Ongoing)

IoT safety prototype under development on Raspberry Pi Pico, integrating MPU6050 IMU, GPS and GSM modules for accident detection, geolocation tagging and automated emergency alerts; follows the literature reviewed in the IECS 2026 Best Paper Award-winning paper.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Web Development: React, Next.js, TypeScript, Firebase, HTML/CSS

Embedded & Robotics: Arduino, Raspberry Pi Pico, ROS2, Sensor Interfacing, Autonomous Navigation

EDA & Simulation: Altium Designer, CST Studio Suite

IoT & Hardware: IoT Prototyping, Hardware Design, Sensor Fusion, Electronics Integration

Research Areas: RF/Microwave Engineering, Antenna Design, Wireless Communication, Autonomous Systems

OS & Tools: Linux, Git, GitHub, Embedded Debugging